

141. The set of all values of $t \in \mathbb{R}$ for which the matrix

$$\begin{bmatrix} e^t & e^{-1}(\sin t - 2\cos t) & e^{-1}(-2\sin t - \cos t) \\ e^t & e^{-1}(2\sin t + \cos t) & e^{-1}(\sin t - 2\cos t) \\ e^t & e^{-1}\cos t & e^{-1}\sin t \end{bmatrix} \text{ is invertible, is}$$

$t \in \mathbb{R}$ के सभी मानों, जिनके लिए आव्यूह

$$\begin{bmatrix} e^t & e^{-1}(\sin t - 2\cos t) & e^{-1}(-2\sin t - \cos t) \\ e^t & e^{-1}(2\sin t + \cos t) & e^{-1}(\sin t - 2\cos t) \\ e^t & e^{-1}\cos t & e^{-1}\sin t \end{bmatrix} \text{ व्युत्क्रमणीय है, का समुच्चय है :}$$

- (A) $\mathbb{R}$ (B) $\{k\pi, k \in \mathbb{Z}\}$ (C) $\left\{(2k+1)\frac{\pi}{2}, k \in \mathbb{Z}\right\}$ (D) $\left\{k\pi + \frac{\pi}{4}, k \in \mathbb{Z}\right\}$

Ans. A

Sol. $|A| = \begin{vmatrix} e^t & e^{-1}(\sin t - 2\cos t) & e^{-1}(-2\sin t - \cos t) \\ e^t & e^{-1}(2\sin t + \cos t) & e^{-1}(\sin t - 2\cos t) \\ e^t & e^{-1}\cos t & e^{-1}\sin t \end{vmatrix}$

$$= e^t \cdot e^{-t} \cdot e^{-t} \begin{vmatrix} 1 & \sin t - 2\cos t & -2\sin t - \cos t \\ 1 & 2\sin t + \cos t & \sin t - 2\cos t \\ 1 & \cos t & \sin t \end{vmatrix}$$

$$R_1 \rightarrow R_1 - R_3, R_2 \rightarrow R_2 - R_3$$

$$|A| = e^{-t} \begin{vmatrix} 0 & \sin t - 3\cos t & -3\sin t - \cos t \\ 0 & 2\sin t & -2\cos t \\ 1 & \cos t & \sin t \end{vmatrix}$$

$$= e^{-t}(-2\cos t(\sin t - 3\cos t) + 2\sin t(3\sin t + \cos t))$$

$$= e^{-t}(6\cos^2 t + 6\sin^2 t) = 6e^{-t}$$

$$|A| \neq 0 \forall t \in \mathbb{R}$$

142. Let $A = \begin{pmatrix} m & n \\ p & q \end{pmatrix}$, $d = |A| \neq 0$ and $|A - d(\text{Adj } A)| = 0$. Then

माना $A = \begin{pmatrix} m & n \\ p & q \end{pmatrix}$, $d = |A| \neq 0$ तथा $|A - d(\text{Adj } A)| = 0$ है। तो :

- (A) $1 + d^2 = (m + q)^2$ (B) $(1 + d)^2 = (m + q)^2$
(C) $(1 + d)^2 = m^2 + q^2$ (D) $1 + d^2 = m^2 + q^2$

Ans. B

Sol. $A = \begin{bmatrix} m & n \\ p & q \end{bmatrix}$

$$d = mq - np$$

$$\text{adj}A = \begin{bmatrix} q & -n \\ -p & m \end{bmatrix}$$

$$|A - d \text{Adj} A| = 0$$

$$\begin{bmatrix} m - dq & n + dn \\ p + dp & q - dm \end{bmatrix}$$

$$(m - dq)(q - dm) - (d + 1)^2 np = 0$$

$$mq - dm^2 - dq^2 + d^2mq = (d + 1)^2 np$$

$$mq(1 + d^2) - d(m^2 + q^2) = (d + 1)^2 (mq - d)$$

$$mq(1 + d^2) - d(m^2 + q^2) = (d^2 + 2d + 1)mq - (d + 1)^2 d$$

$$(m + q)^2 = (d + 1)^2$$

143. If P is 3×3 real matrix such that $P^T = aP + (a - 1)I$, where $a > 1$, then

(A) $|\text{Adj} P| > 1$

(B) $|\text{Adj} P| = \frac{1}{2}$

(C) $|\text{Adj} P| = 1$

(D) P is a singular matrix

यदि P एक 3×3 का वास्तविक आव्यूह इस प्रकार है कि $P^T = aP + (a - 1)I$ है, जहाँ $a > 1$ है, तब :

(A) $|\text{Adj} P| > 1$

(B) $|\text{Adj} P| = \frac{1}{2}$

(C) $|\text{Adj} P| = 1$

(D) P एक singular matrix है

Ans. C

Sol. $(P^T)^T = (aP + (a - 1)I)^T$

$$\Rightarrow P = aP^T + (a - 1)I$$

$$\Rightarrow P = a(ap + (a - 1)I) + (a - 1)I$$

$$\Rightarrow P = a^2P + a(a - 1)I + (a - 1)I$$

$$\Rightarrow P = a^2P + (a^2 - 1)I$$

$$\Rightarrow (a^2 - 1)P + (a^2 - 1)I = 0$$

$$\Rightarrow (a^2 - 1)(P + I) = 0$$

$$\therefore a > 1$$

$$\Rightarrow P + I = 0$$

$$\Rightarrow P = -I$$

$$\text{adj} P = \text{adj}(-I)$$

$$\Rightarrow |\text{adj} P| = |\text{adj}(-I)| = |-I|^2$$

$$\Rightarrow |\text{adj} P| = 1$$

144. Let $A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 4 & -1 \\ 0 & 12 & -3 \end{pmatrix}$. Then the sum of the diagonal elements of the matrix $(A + I)^{11}$ is equal to :

माना $A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 4 & -1 \\ 0 & 12 & -3 \end{pmatrix}$ है। तो आव्यूह $(A + I)^{11}$ के विकर्ण के अवयवों का योग है :

(A) 6144

(B) 4094

(C) 2050

(D) 4097

Ans. D

Sol. $A^2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 4 & -1 \\ 0 & 12 & -3 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 4 & -1 \\ 0 & 12 & -3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 4 & -1 \\ 0 & 12 & -3 \end{bmatrix} = A$

Now $(A + I)^{11} = {}^{11}C_0 A^{11} + {}^{11}C_1 A^{10} + \dots + {}^{11}C_{11} I$
 $= A ({}^{11}C_0 + {}^{11}C_1 + \dots + {}^{11}C_{10}) + I$
 $= A (2^{11} - 1) + I$

Trace of $(A + I)^{11}$

$= 2^{11} + 4(2^{11} - 1) + 1 + (-3)(2^{11} - 1) + 1$

$= 2 \times 2^{11} + 1$

$= 2^{12} + 1$

145. Let $A = [a_{ij}]_{2 \times 2}$, where $a_{ij} \neq 0$ for all i, j and $A^2 = I$. Let a be the sum of all diagonal elements of A and $b = |A|$. Then $3a^2 + 4b^2$ is equal to :

माना $A = [a_{ij}]_{2 \times 2}$, जहाँ सभी i, j के लिए $a_{ij} \neq 0$ एवं $A^2 = I$ है। माना A के विकर्ण के सभी अवयवों का योग a है और $b = |A|$ है।

तब $3a^2 + 4b^2$ बराबर है :

(A) 7

(B) 3

(C) 14

(D) 4

Ans. D

Sol. Let $A = \begin{bmatrix} p & q \\ r & s \end{bmatrix}$

$A^2 = \begin{bmatrix} p & q \\ r & s \end{bmatrix} \begin{bmatrix} p & q \\ r & s \end{bmatrix} = \begin{bmatrix} p^2 + qr & pq + qs \\ rp + rs & qr + s^2 \end{bmatrix}$

$A^2 = I$

$\Rightarrow p^2 + qr = 1 \quad q(p + s) = 0$

$r(p + s) = 0 \quad qr + s^2 = 1$

$$q \neq 0 \Rightarrow p + s = 0 \Rightarrow a = 0$$

$$b = |A| = ps - qr = -p^2 - qr = -1 (\because s = -p)$$

$$\therefore 3a^2 + 4b^2 = 4$$

- 146.** Let P be a square matrix such that $P^2 = I - P$. For $\alpha, \beta, \gamma, \delta \in \mathbb{N}$, if $P^\alpha + P^\beta = \gamma I - 29P$ and $P^\alpha - P^\beta = \delta I - 13P$, then $\alpha + \beta + \gamma - \delta$ is equal to :

माना P एक वर्ग आव्यूह है जिसके लिए $P^2 = I - P$ है। तो $\alpha, \beta, \gamma, \delta \in \mathbb{N}$ के लिए यदि $P^\alpha + P^\beta = \gamma I - 29P$ तथा $P^\alpha - P^\beta = \delta I - 13P$ है, तो $\alpha + \beta + \gamma - \delta$ बराबर है :

- (A) 18 (B) 22 (C) 40 (D) 24

Ans. D

Sol.

$$P^2 = I - P$$

$$P^\alpha + P^\beta = \gamma I - 29P$$

$$P^\alpha - P^\beta = \delta I - 13P$$

$$P^4 = (I - P)^2 = I + P^2 - 2P$$

$$P^4 = I + I - P - 2P = 2I - 3P$$

$$P^8 = (P^4)^2 = (2I - 3P)^2 = 4I + 9P^2 - 12P$$

$$= 4I + 9(I - P) - 12P$$

$$P^8 = 13I - 21P$$

$$\begin{aligned} P^6 &= P^4 \cdot P^2 &= (2I - 3P)(I - P) \\ & &= 2I - 5P + 3P^2 \\ & &= 2I - 5P + 3(I - P) \\ & &= 5I - 8P \end{aligned}$$

$$(1) + (2)$$

$$P^8 + P^6 = 18I - 29P$$

From (A)

$$\alpha = 8, \quad \beta = 6$$

$$\gamma = 18$$

$$\delta = 8$$

$$\alpha + \beta + \gamma - \delta = 32 - 8 = 24$$

$$(1) - (2)$$

$$P^8 - P^6 = 8I - 13P$$

- 147.** If $A = \frac{1}{5!6!7!} \begin{bmatrix} 5! & 6! & 7! \\ 6! & 7! & 8! \\ 7! & 8! & 9! \end{bmatrix}$, then $|\text{adj}(\text{adj}(2A))|$ is equal to :

यदि $A = \frac{1}{5!6!7!} \begin{bmatrix} 5! & 6! & 7! \\ 6! & 7! & 8! \\ 7! & 8! & 9! \end{bmatrix}$ है, तो $|\text{adj}(\text{adj}(2A))|$ बराबर है :

(A) 2^{16}

(B) 2^8

(C) 2^{20}

(D) 2^{12}

Ans. A

Sol. $|A| = \frac{1}{5!6!7!} \times 5!6!7! \begin{vmatrix} 1 & 6 & 42 \\ 1 & 7 & 56 \\ 1 & 8 & 72 \end{vmatrix}$

$$R_2 \rightarrow R_2 - R_1 \text{ \& } R_3 \rightarrow R_3 - R_1$$

$$|A| = \begin{vmatrix} 1 & 6 & 42 \\ 0 & 1 & 14 \\ 0 & 1 & 16 \end{vmatrix} \Rightarrow |A| = 2$$

$$|\text{Adj}(\text{adj}(2A))| = |2A|^{(n-1)^2} = |2A|^4 = 2^{12} |A|^4 = 2^{16}$$

148. Let A be a 2×2 matrix with real entries such that $A' = \alpha A + I$, where $\alpha \in \mathbb{R} - \{-1, 1\}$. If $\det(A^2 - A) = 4$, then the sum of all possible values of α is equal to :

माना A वास्तविक अवयवों का एक 2×2 आव्यूह है जिसके लिए $A' = \alpha A + I$ है, $\alpha \in \mathbb{R} - \{-1, 1\}$ है। यदि $\det(A^2 - A) = 4$ है, तो α के सभी संभव मानों का योग बराबर है :

(A) $\frac{3}{2}$

(B) 2

(C) $\frac{5}{2}$

(D) 0

Ans. C

Sol. $A^T = \alpha A + I$
 $A = \alpha A^T + I$
 $A = \alpha(\alpha A + I) + I$
 $A = \alpha^2 A + (\alpha + 1)I$
 $A(1 - \alpha^2) = (\alpha + 1)I$
 $A = \frac{I}{1 - \alpha} \dots (1)$
 $|A| = \frac{1}{(1 - \alpha)^2} \dots (2)$
 $|A^2 - A| = |A| |A - I| \dots (3)$

$$A - I = \frac{I}{1 - \alpha} - I = \frac{\alpha}{1 - \alpha} I$$

$$|A - I| = \left(\frac{\alpha}{1 - \alpha} \right)^2 \dots (4)$$

$$\text{Now } |A^2 - A| = 4$$

$$|A| |A - I| = 4$$

$$\Rightarrow \frac{1}{(1 - \alpha)^2} \alpha^2 = 4$$

$$\Rightarrow \frac{\alpha}{(1 - \alpha)^2} = \pm 2$$

$$\Rightarrow 2(1 - \alpha)^2 = \pm \alpha$$

$$(C_1) 2(1 - \alpha)^2 = \alpha$$

$$(C_2) 2(1 - \alpha)^3 = -\alpha$$

$$2\alpha^2 - 5\alpha + 2 = 0 \begin{cases} \alpha_1 \\ \alpha_2 \end{cases}$$

$$2\alpha^2 - 3\alpha + 2 = 0$$

$$\alpha_1 + \alpha_2 = \frac{5}{2}$$

$$\alpha \notin \mathbb{R}$$

- 149.** Let $S = \{M = [a_{ij}], a_{ij} \in \{0, 1, 2\}, 1 \leq i, j \leq 2\}$ be a sample space and $A = \{M \in S : M \text{ is invertible}\}$ be an event. Then $P(A)$ is equal to :

माना $S = \{M = [a_{ij}], a_{ij} \in \{0, 1, 2\}, 1 \leq i, j \leq 2\}$ एक प्रतिदर्श समष्टि है तथा $A = \{M \in S : M \text{ व्युत्क्रमणीय है}\}$ एक घटना है। तो $P(A)$ बराबर है :

(A) $\frac{50}{81}$

(B) $\frac{47}{81}$

(C) $\frac{49}{81}$

(D) $\frac{16}{27}$

Ans. A

Sol. $M = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, where $a, b, c, d \in \{0, 1, 2\}$

$$n(s) = 3^4 = 81$$

we first find $p(\bar{A})$

$$|m| = 0 \Rightarrow ad = bc$$

$$ad = bc = 0 \Rightarrow \text{no. of } (a, b, c, d) = (3^2 - 2^2)^2 = 25$$

$$ad = bc = 1 \Rightarrow \text{no. of } (a, b, c, d) = 1^2 = 1$$

$$ad = bc = 2 \Rightarrow \text{no. of } (a, b, c, d) = 2^2 = 4$$

$$ad = bc = 4 \Rightarrow \text{no. of } (a, b, c, d) = 1^2 = 1$$

$$\therefore P(\bar{A}) = \frac{31}{81} \Rightarrow P(A) = \frac{50}{81}$$

150. Let $A = \begin{bmatrix} 1 & \frac{1}{51} \\ 0 & 1 \end{bmatrix}$. If $B = \begin{bmatrix} 1 & 2 \\ -1 & -1 \end{bmatrix} A \begin{bmatrix} -1 & -2 \\ 1 & 1 \end{bmatrix}$, then the sum of all the elements of the matrix $\sum_{n=1}^{50} B^n$ is equal to

:

माना $A = \begin{bmatrix} 1 & \frac{1}{51} \\ 0 & 1 \end{bmatrix}$ है। यदि $B = \begin{bmatrix} 1 & 2 \\ -1 & -1 \end{bmatrix} A \begin{bmatrix} -1 & -2 \\ 1 & 1 \end{bmatrix}$ है, तो $\sum_{n=1}^{50} B^n$ के सभी अवयवों का योग है :

(A) 75

(B) 100

(C) 50

(D) 125

Ans. B

Sol. $\therefore \begin{bmatrix} 1 & 2 \\ -1 & -1 \end{bmatrix}^{(M)} \begin{bmatrix} -1 & -2 \\ 1 & 1 \end{bmatrix}^{(N)} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

$$\therefore MN = I = NM$$

$$B = MAN$$

$$B^n = (MAN)^n = (MAN)(MAN)\dots$$

$$= (MA^2N)MAN$$

$$= MA^nN$$

$$\text{Now } A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} 0 & \frac{1}{51} \\ 0 & 0 \end{bmatrix} = I + E$$

$$E^2 = 0$$

$$\therefore A^n = (I + E)^n = I + nE + \underbrace{{}^nC_2 E^2}_0$$

$$= I + nE$$

$$= \begin{bmatrix} 1 & \frac{n}{51} \\ 0 & 1 \end{bmatrix}$$

$$\text{So, } B^n = MA^nN = \begin{bmatrix} 1 + \frac{n}{51} & \frac{n}{51} \\ \frac{-n}{51} & 1 - \frac{n}{51} \end{bmatrix}$$

$$\sum_{n=1}^{50} B^n = \begin{bmatrix} 50 + \frac{50 \cdot 51}{2 \cdot 51} & \frac{50 \cdot 51}{2 \cdot 51} \\ \frac{-50 \cdot 51}{2 \cdot 51} & 50 - \frac{50 \cdot 51}{2 \cdot 51} \end{bmatrix} = \begin{bmatrix} 75 & 25 \\ -25 & 25 \end{bmatrix}$$

$$\therefore \text{sum} = 100$$

151. The number of symmetric matrices of order 3, with all the entries from the set $\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$, is:

कोटि 3 के सममित आव्यूहों, जिनके सभी अवयव समुच्चय $\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$ से हैं, की संख्या है :

- (A) 6^{10} (B) 10^6 (C) 9^{10} (D) 10^9

Ans. B

Sol. $A = \begin{bmatrix} a & b & c \\ b & d & e \\ c & e & f \end{bmatrix}, a, b, c, d, e, f \in \{0, 1, 2, \dots, 9\}$

$$\text{Number of matrices} = 10^6$$

152. Let $B = \begin{bmatrix} 1 & 3 & \alpha \\ 1 & 2 & 3 \\ \alpha & \alpha & 4 \end{bmatrix}$, $\alpha > 2$ be the adjoint of a matrix A and $|A| = 2$. Then $[\alpha - 2\alpha \ \alpha]B \begin{bmatrix} \alpha \\ -2\alpha \\ \alpha \end{bmatrix}$ is equal to :

माना $B = \begin{bmatrix} 1 & 3 & \alpha \\ 1 & 2 & 3 \\ \alpha & \alpha & 4 \end{bmatrix}$, $\alpha > 2$, एक आव्यूह A का सहखण्डज है तथा $|A| = 2$ है। तो $[\alpha - 2\alpha \ \alpha]B \begin{bmatrix} \alpha \\ -2\alpha \\ \alpha \end{bmatrix}$ बराबर है :

- (A) -16 (B) 16 (C) 0 (D) 32

Ans. A

Sol. $|B| = |\text{adj } A| = |A|^{3-1} = |A|^2 = 4$

$$1(8 - 3\alpha) - 3(4 - 3\alpha) + \alpha(\alpha - 2\alpha) = 4$$

$$\Rightarrow 8 - 3\alpha - 12 + 9\alpha - \alpha^2 = 4$$

$$\Rightarrow \alpha^2 - 6\alpha + 8 = 0$$

$$\Rightarrow \alpha = 2, 4$$

$$\Rightarrow \alpha = 4 \quad (\because \alpha > 2)$$

Now

$$\begin{bmatrix} 4 & -8 & 4 \end{bmatrix} \begin{bmatrix} 1 & 3 & 4 \\ 1 & 2 & 3 \\ 4 & 4 & 4 \end{bmatrix} \begin{bmatrix} 4 \\ -8 \\ 4 \end{bmatrix}$$

$$= 16 \begin{bmatrix} 1 & -2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 3 & 4 \\ 1 & 2 & 3 \\ 4 & 4 & 4 \end{bmatrix} \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}$$

$$= 16 \begin{bmatrix} 1-2+4 & 3-4+4 & 4-6+4 \end{bmatrix} \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}$$

$$= 16 [3 - 6 + 2]$$

$$= 16 [-1] = [-16]$$

- 153.** Let the determinant of a square matrix A of order m be $m - n$, where m and n satisfy $4m + n = 22$ and $17m + 4n = 93$. If $\det(n \operatorname{adj}(\operatorname{adj}(mA))) = 3^a 5^b 6^c$, then $a + b + c$ is equal to :

यदि m कोटि के वर्ग आव्यूह A के सारणिक का मान $m - n$ है, जहाँ m तथा n, $4m + n = 22$ तथा $17m + 4n = 93$ को संतुष्ट करते हैं। यदि $\det(n \operatorname{adj}(\operatorname{adj}(mA))) = 3^a 5^b 6^c$ हो, तो $a + b + c$ के मान बराबर होगा :

- (A) 96 (B) 101 (C) 109 (D) 84

Ans. A

Sol. $|A| = m - n$

$$4m + n = 22$$

$$17m + 4n = 93$$

$$m = 5, n = 2$$

$$|A| = 3$$

$$|2 \operatorname{adj}(\operatorname{adj} 5A)| = 2^5 |5A|^{16}$$

$$= 2^5 \cdot 5^{80} |A|^{16}$$

$$= 2^5 \cdot 5^{80} \cdot 3^{16}$$

$$= 3^{11} \cdot 5^{80} \cdot 6^5$$

$$a + b + c = 96$$

- 154.** Let $P = \begin{bmatrix} 3 & -1 & -2 \\ 2 & 0 & \alpha \\ 3 & -5 & 0 \end{bmatrix}$, where $\alpha \in \mathbb{R}$. Suppose $Q = [q_{ij}]$ is a matrix satisfying $PQ = kI_3$ for some non-zero $k \in \mathbb{R}$.

If $q_{23} = -\frac{k}{8}$ and $|Q| = \frac{k^2}{2}$ then $\alpha^2 + k^2$ is equal to _____.

माना $P = \begin{bmatrix} 3 & -1 & -2 \\ 2 & 0 & \alpha \\ 3 & -5 & 0 \end{bmatrix}$ है, जबकि $\alpha \in \mathbb{R}$ है। माना $Q = [q_{ij}]$ एक आव्यूह है, जिसके लिए $PQ = kI_3$, किसी शून्येतर, $k \in \mathbb{R}$

के लिए, है। यदि $q_{23} = -\frac{k}{8}$ तथा $|Q| = \frac{k^2}{2}$ हैं, तो $\alpha^2 + k^2$ बराबर है _____।

Ans. 17

Sol. $P = \begin{bmatrix} 3 & -1 & -2 \\ 2 & 0 & \alpha \\ 3 & -5 & 0 \end{bmatrix}$

$$|P| = 4(3\alpha + 5)$$

$$PQ = kI_3$$

$$|P| \cdot |Q| = k^3$$

$$4(3\alpha + 5) \cdot \frac{k^2}{2} = k^3$$

$$k = 2(3\alpha + 5) \dots (1)$$

$$Q = kP^{-1}$$

$$Q = \frac{k \text{ adj. } P}{|P|}$$

$$q_{23} = \frac{k}{4(3\alpha + 5)} \cdot C_{32} \quad (C_{ij} = \text{Cofactor of } a_{ij})$$

$$\frac{-k}{8} = \frac{k}{4(3\alpha + 5)} \times (-1) \times (3\alpha + 6)$$

$$\alpha = -1$$

$$\alpha^2 + k^2 = 17$$

155. Let $A = \begin{bmatrix} x & y & z \\ y & z & x \\ z & x & y \end{bmatrix}$, where x, y and z are real numbers such that $x + y + z > 0$ and $xyz = 2$. If $A^2 = I_3$, then the value of $x^3 + y^3 + z^3$ is _____.

माना $A = \begin{bmatrix} x & y & z \\ y & z & x \\ z & x & y \end{bmatrix}$ है, जहाँ x, y तथा z वास्तविक संख्याएँ हैं, जिनके लिए $x + y + z > 0$ तथा $xyz = 2$ हैं।

यदि $A^2 = I_3$ है, तो $x^3 + y^3 + z^3$ का मान है :

Ans. 7

Sol.
$$\begin{bmatrix} x & y & z \\ y & z & x \\ z & x & y \end{bmatrix} \begin{bmatrix} x & y & z \\ y & z & x \\ z & x & y \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} x^2 + y^2 + z^2 & xy + yz + zx & xy + yz + zx \\ xy + yz + zx & x^2 + y^2 + z^2 & xy + yz + zx \\ xy + yz + zx & xy + yz + zx & x^2 + y^2 + z^2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$x^2 + y^2 + z^2 = 1$$

$$xy + yz + zx = 0$$

$$(x + y + z)^2 = \sum x^2 + 2 \sum xy$$

$$(x + y + z)^2 = 1 + 0$$

$$x + y + z = \pm 1$$

$$\Rightarrow x + y + z = 1$$

$$x^3 + y^3 + z^3 = x^3 + y^3 + z^3 - 3xyz + 3xyz$$

$$= (x + y + z)(x^2 + y^2 + z^2 - xy - yz - zx) + 3xyz$$

$$= (1)(1) + 3 \times 2$$

$$= 7$$

156. If the matrix $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 3 & 0 & -1 \end{bmatrix}$ satisfies the equation $A^{20} + \alpha A^{19} + \beta A$ $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ for some real numbers α

and β , then $\beta - \alpha$ is equal to _____.

यदि किसी वास्तविक संख्याओं α तथा β के लिए आव्यूह $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 3 & 0 & -1 \end{bmatrix}$, समीकरण

$$A^{20} + \alpha A^{19} + \beta A \begin{bmatrix} 1 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 1 \end{bmatrix} \text{ को सन्तुष्ट करता है, तो } \beta - \alpha \text{ बराबर है } \underline{\hspace{2cm}} \text{।}$$

Ans. 4

Sol. $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 3 & 0 & -1 \end{bmatrix}$

$$A^2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 3 & 0 \\ 3 & 0 & -1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 3 & 0 & -1 \end{bmatrix}$$

$$A^2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2^2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$A^3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2^3 & 0 \\ 3 & 0 & -1 \end{bmatrix}$$

$$A^4 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2^4 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

By observing the pattern

$$A^{20} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2^{20} & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$A^{19} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2^{19} & 0 \\ 3 & 0 & -1 \end{bmatrix}$$

$$A^{20} + \alpha A^{19} + \beta A = \begin{bmatrix} 1 + \alpha + \beta & 0 & 0 \\ 0 & 2^{20} + \alpha \cdot 2^{19} + 2\beta & 0 \\ 3\alpha + 3\beta & 0 & 1 - \alpha - \beta \end{bmatrix}$$

$$\alpha + \beta = 0 \text{ and } 2^{20} + \alpha \cdot 2^{19} + 2\beta = 4$$

$$2^{20} + \alpha \cdot 2^{19} - 2\alpha = 4$$

$$\Rightarrow 2^{19}(\alpha + 2) = 2(\alpha + 2)$$

$$(\alpha + 2)(2^{19} - 2) = 0$$

$$\alpha = -2$$

$$\therefore \beta = 2$$

$$\therefore |\alpha - \beta| = |-2 - 2| = |4|$$

$$\beta - \alpha = 2 - (-2) = 4.$$

- 157.** If $A = \begin{bmatrix} 2 & 3 \\ 0 & -1 \end{bmatrix}$, then the value of $\det(A^4) + \det(A^{10} - (\text{Adj}(2A))^{10})$ is equal to :

यदि $A = \begin{bmatrix} 2 & 3 \\ 0 & -1 \end{bmatrix}$ है, तो $\det(A^4) + \det(A^{10} - (\text{Adj}(2A))^{10})$ का मान बराबर है :

Ans. 16

Sol. $|A| = -2$

$$|A|^4 = 16$$

$$A^{10} = \begin{bmatrix} 2^{10} & 2^{10} - 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1024 & 1023 \\ 0 & 1 \end{bmatrix}$$

$$2A = \begin{bmatrix} 4 & 6 \\ 0 & -2 \end{bmatrix}$$

$$\text{adj}(2A) = \begin{bmatrix} -2 & -6 \\ 0 & 4 \end{bmatrix}$$

$$(\text{adj}(2A))^{10} = 2^{10} \begin{bmatrix} 1 & 3 \\ 0 & -2 \end{bmatrix}^{10}$$

$$= 2^{10} \begin{bmatrix} 1 & -(2^{10} - 1) \\ 0 & 2^{10} \end{bmatrix}$$

$$A^{10} - (\text{adj}(2A))^{10} = \begin{bmatrix} 0 & 2^{11} \times 1023 \\ 0 & 1 - (1024)^2 \end{bmatrix}$$

$$|A^{10} - (\text{adj}(2a))^{10}| = 0$$

- 158.** Let $A = \begin{bmatrix} a_1 \\ a_2 \end{bmatrix}$ and $B = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$ be two 2×1 matrices with real entries such that $A = XB$, where $X = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 & -1 \\ 1 & k \end{bmatrix}$ and $k \in \mathbb{R}$. If $a_1^2 + a_2^2 = \frac{2}{3}(b_1^2 + b_2^2)$ and $(k^2 + 1)b_2^2 \neq -2b_1b_2$ then the value of k is _____.

माना 2×1 के दो आव्यूह $A = \begin{bmatrix} a_1 \\ a_2 \end{bmatrix}$ तथा $B = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$ हैं जिनके अवयव वास्तविक हैं तथा $A = XB$ है, जहाँ $X = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 & -1 \\ 1 & k \end{bmatrix}$

और $k \in \mathbb{R}$ है। यदि $a_1^2 + a_2^2 = \frac{2}{3}(b_1^2 + b_2^2)$ तथा $(k^2 + 1)b_2^2 \neq -2b_1b_2$ हैं, तो k का मान है _____।

Ans. 1

Sol. $A = XB$

$$\begin{bmatrix} a_1 \\ a_2 \end{bmatrix} = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 & -1 \\ 1 & k \end{bmatrix} \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$$

$$a_1 = \frac{1}{\sqrt{3}}(b_1 - b_2)$$

$$a_2 = \frac{1}{\sqrt{3}}(b_1 + kb_2)$$

$$a_1^2 + a_2^2 = \frac{1}{3} \left\{ (b_1 - b_2)^2 + (b_1 + kb_2)^2 \right\}$$

$$\frac{2}{3}(b_1^2 + b_2^2) = \frac{1}{3} \left\{ (b_1 - b_2)^2 + (b_1 + kb_2)^2 \right\}$$

$$2(b_1^2 + b_2^2) = \left\{ (b_1 - b_2)^2 + (b_1 + kb_2)^2 \right\}$$

$$k = 1$$

- 159.** Let $P = \begin{bmatrix} -30 & 20 & 56 \\ 90 & 140 & 112 \\ 120 & 60 & 14 \end{bmatrix}$ and $A = \begin{bmatrix} 2 & 7 & \omega^2 \\ -1 & -\omega & 1 \\ 0 & -\omega & \omega + 1 \end{bmatrix}$ where $\omega = \frac{-1 + i\sqrt{3}}{2}$, and I_3 be the identity matrix of

order 3. If the determinant of the matrix $(P^{-1}AP - I_3)^2$ is $\alpha\omega^2$, then the value of α is equal to _____.

माना $P = \begin{bmatrix} -30 & 20 & 56 \\ 90 & 140 & 112 \\ 120 & 60 & 14 \end{bmatrix}$ तथा $A = \begin{bmatrix} 2 & 7 & \omega^2 \\ -1 & -\omega & 1 \\ 0 & -\omega & \omega + 1 \end{bmatrix}$ है, जहाँ $\omega = \frac{-1 + i\sqrt{3}}{2}$ है तथा 3 कोटी का तत्समक आव्यूह

I_3 है। यदि आव्यूह $(P^{-1}AP - I_3)^2$ का सारणिक $\alpha\omega^2$ है, तो α का मान बराबर है _____ ।

Ans. 36

Sol. $M = (P^{-1}AP - I)(P^{-1}AP - I)$

$$= P^{-1}AP P^{-1}AP - P^{-1}AP - P^{-1}AP + I$$

$$= P^{-1}A^2P - 2P^{-1}AP + P^{-1}IP$$

$$= P^{-1}(A^2 - 2A + I)P$$

$$M = P^{-1}(A - I)^2P$$

$$|M| = |P^{-1}| |A - I|^2 |P|$$

$$= |A - I|^2$$

$$|A - I|^2 = \begin{vmatrix} 1 & 7 & \omega^2 \\ -1 & -1-\omega & 1 \\ 0 & -\omega & -\omega \end{vmatrix}^2$$

$$= \begin{vmatrix} 1 & 7 & \omega^2 \\ -1 & \omega^2 & 1 \\ 0 & -\omega & -\omega \end{vmatrix}^2$$

$$= (-1 + \omega + 7(-\omega) + \omega^2(\omega))^2$$

$$= (-6\omega)^2 = 36\omega^2$$

$$\Rightarrow \alpha = 36$$

160. The total number of 3×3 matrices A having entries from the set $\{0, 1, 2, 3\}$ such that the sum of all the diagonal entries of AA^T is 9, is equal to _____.

3×3 के आव्यूहों A, जिनके अवयव समुच्चय $\{0, 1, 2, 3\}$ में से हैं तथा AA^T के विकर्ण के सभी अवयवों का योगफल 9 है, की कुल संख्या है _____ ।

Ans. 766

Sol. $A = \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix}$

Sum of diagonal entries of $AA^T = \sum a_i^2 + \sum b_i^2 + \sum c_i^2$

$$\{0, 0, 0, \dots, 0, 3\} \Rightarrow \frac{9}{3} = 9$$

$$\{0, 0, 0, 1, 1, 1, 1, 1, 2\} \Rightarrow \frac{9}{3 \cdot 5 \cdot 11} = 504$$

$$\{2, 2, 1, 0, 0, 0, 0, 0, 0\} \Rightarrow \frac{9}{2 \cdot 6} = 252$$

$$\{1, 1, 1, 1, 1, 1, 1, 1, 1\} \Rightarrow \frac{9}{9} = 1$$

$$\text{Total} = 9 + 504 + 252 + 1 = 766$$